

Doppler-Enhanced Learning-Free Multi-Object Tracking using 4D Radar

Yuhe Wen, Jaeho Jung, Quan Gan, Jehwan Choi, Duy-Linh Nguyen, Xuan-Thuy Vo, Kang-Hyun Jo
Dept. of Electrical, Electronic and Computer Engineering, University of Ulsan, Ulsan, Korea
{wenyuhe, jjho0514, ganquan, ndlinh301, blackhammer5}@mail.ulsan.ac.kr; {cjh1897, acejo}@ulsan.ac.kr

Abstract—Four-dimensional (4D) imaging radar provides radial velocity (Doppler), radar cross section (RCS), and detection confidence alongside 3D spatial information. Yet among existing learning-free multi-object trackers, these signals remain largely unexploited. This paper proposes DE-FastPoly, a plug-in extension of the FastPoly tracker with three core training-free modules—Doppler velocity initialization (DVI), measured- v_r Doppler association (DA), and score-adaptive measurement noise (SAN)—plus an optional RCS-aware association module (RCSA) as a dataset-specific extension. On the View-of-Delft (VoD) benchmark, DE-FastPoly achieves MOTA 0.361 under realistic detection (+6.7 pp (percentage points) over the FastPoly baseline) and MOTA 0.954 in the oracle setting (+11.7 pp), while running at 217 FPS with modest computational overhead (+6.3%). Cross-dataset transfer on nuScenes with a DA+SAN configuration reduces identity switches (IDS) by 16% without degrading AMOTA. All modules are learning-free and require no retraining.

Index Terms—4D imaging radar, autonomous driving, Doppler velocity, Kalman filter, multi-object tracking

I. INTRODUCTION

Multi-object tracking (MOT) in autonomous driving requires accurate localization and consistent identity maintenance of dynamic agents across sensor frames. The tracking-by-detection paradigm, in which a detector produces per-frame bounding boxes and a tracker associates them across time, has proven effective in both 2D camera and 3D LiDAR settings [1], [2]. However, the introduction of 4D imaging radar as a primary sensor creates new opportunities. Unlike LiDAR, 4D radar provides a direct radial velocity (Doppler) measurement for each detection, as well as radar cross section (RCS) and detection confidence, at significantly lower cost and with greater robustness to weather and lighting conditions [3], [4].

Despite this, existing learning-free 3D trackers such as FastPoly [5] and SimpleTrack [2] rely primarily on geometric cues for data association and track state estimation, and do not explicitly exploit the accompanying Doppler or RCS attributes in the tracking pipeline. This leaves potentially valuable motion and appearance information unused.

This paper proposes DE-FastPoly, which integrates four lightweight, training-free modules into the FastPoly tracker to exploit radar-specific cues at three stages of the predict-associate-update cycle:

- **Doppler velocity initialization (DVI):** radial velocity is used to seed the Kalman filter velocity state for newly created tracks, reducing convergence latency.

- **Measured- v_r Doppler association (DA):** a velocity-consistency soft term is blended into the association cost using the stored measured v_r from each track's last matched detection.
- **Score-adaptive measurement noise (SAN):** detection confidence is used to scale the Kalman measurement noise covariance, allowing high-confidence detections to update the state more aggressively.
- **RCS-aware association (RCSA):** radar cross section consistency is used as a training-free, class-specific soft cue for pedestrian association.

All modules are learning-free additions that require no labeled data, no retraining, and no additional changes to the detection pipeline. Evaluation is conducted on the View-of-Delft (VoD) 4D radar dataset, with a cross-dataset transfer study on nuScenes. The proposed method achieves consistent MOTA improvements and identity switch (IDS) reductions under both oracle and realistic detection settings, and maintains real-time throughput (217 frames per second, FPS).

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the four proposed modules. Section IV presents experimental results and ablation studies. Section V concludes the paper and outlines future directions.

II. RELATED WORK

A. 3D Multi-Object Tracking and Doppler-Aware Methods

The tracking-by-detection paradigm has been widely adopted for 3D MOT in autonomous driving [1], [2]. Early methods such as AB3DMOT [1] apply Kalman filtering and Hungarian matching with IoU-based cost in 3D space. SimpleTrack [2] introduces Gaussian-to-Box matching and a two-stage association strategy. FastPoly [5] extends this with polynomial motion models for improved prediction. PolyMOT [6] extends polynomial motion modeling to multi-class 3D tracking with class-specific motion models. These methods are designed primarily for LiDAR detections and do not exploit radar-specific attributes among the methods compared here. ByteTrack [7], originating in 2D image-based MOT, is included as a representative baseline in its 3D-adapted form. Several works have explored using Doppler velocity for tracking with radar sensors. RaTrack [8] proposes a joint detection and tracking framework for 4D radar that leverages Doppler in a learned feature representation. Scheiner et al. [9] use

Doppler to filter static radar points and improve the detection quality fed to the tracker. The proposed approach targets a learning-free, plug-in setting. Rather than redesigning the tracking architecture or training a model, Doppler information is incorporated into an existing strong tracker as soft cues that improve both initialization and association.

B. 4D Imaging Radar for Perception

4D imaging radar delivers a denser point cloud than traditional 3D radar and provides additional attributes including Doppler, RCS, and elevation [3], [4]. VoD [3] is a publicly available benchmark for 4D radar-based perception in urban driving, supporting tasks including object detection and MOT. Recent work has demonstrated competitive object detection performance using 4D radar point clouds [3], [4], motivating the development of tracking methods that exploit radar-native attributes beyond position.

III. PROPOSED METHOD

FastPoly [5] is extended with four lightweight, training-free modules that incorporate radar-derived cues into the tracking pipeline. Fig. 1 illustrates the overall architecture. The original predict-associate-update cycle of the base tracker is preserved, and the proposed modules act as plug-in components that enhance three stages of the pipeline: track initialization, data association, and measurement modeling. Fig. 2 shows the per-class radial velocity distribution on VoD, motivating the use of Doppler as a discriminative association cue.

A. Doppler Velocity Initialization (DVI)

Standard Kalman-filter-based trackers initialize the velocity of newly created tracks to zero and rely on subsequent observations to gradually converge to the true motion state.

In 4D radar sensing, each detection provides a radial velocity measurement v_r , representing the projection of object velocity $\mathbf{v}$ onto the sensor line-of-sight:

$$v_r = \mathbf{v} \cdot \hat{\mathbf{r}}, \quad \hat{\mathbf{r}} = \frac{(x, y)^T}{\sqrt{x^2 + y^2}}, \quad r = \sqrt{x^2 + y^2} \quad (1)$$

where (x, y) are the detection's horizontal coordinates in the sensor frame and r its range.

Assuming that the initial motion direction approximately aligns with the radial direction, Cartesian velocity components are initialized as:

$$v_x^{(0)} = v_r \cdot \frac{x}{r}, \quad v_y^{(0)} = v_r \cdot \frac{y}{r} \quad (2)$$

This approximation is commonly adopted when only radial velocity is available at track birth; subsequent Kalman updates quickly correct any initialization error. This provides a physically grounded initial velocity estimate and reduces convergence time during early tracking.

B. Measured- v_r Doppler Association (DA)

The association stage assigns detections to existing tracks by minimizing a cost matrix. Standard geometric costs (IoU, Euclidean distance) do not exploit Doppler information. The cost is augmented with a velocity consistency term derived from raw radar measurements.

Track velocity storage. Each track maintains v_r^{trk} , the radar-measured v_r from its most recently matched detection. A critical design choice is to use the *measured* v_r rather than a Kalman-filter-derived estimate. The filter state velocity $(\hat{v}_x, \hat{v}_y)$ is a smoothed derivative of position innovations; without direct velocity observations it diverges from the true object velocity, and empirically the stored measurement provides more stable associations than projecting the Kalman-filter velocity to the radial direction (see Section IV-C). With `max_age` = 2, unmatched tracks are terminated within two frames, limiting the use of stale velocity measurements in subsequent association steps.

Doppler consistency score. For each detection-track pair where both carry valid v_r , the Doppler consistency score s_d is computed as:

$$s_d = \exp\left(-\frac{(\Delta v_r)^2}{2\sigma^2}\right), \quad \Delta v_r = v_r^{\text{det}} - v_r^{\text{trk}} \quad (3)$$

with $\sigma = 10.0$ m/s, selected empirically based on the typical velocity variation range in VoD and evaluated in the sensitivity study (Section IV-D).

Blended cost. Here c_{geo} denotes the base geometric association cost in the FastPoly pipeline. The blended cost c is:

$$c = \alpha \cdot c_{\text{geo}} + \beta \cdot (1 - s_d), \quad \alpha = 0.95, \quad \beta = 0.05 \quad (4)$$

The weight β is intentionally small to ensure that spatial proximity remains the dominant association cue, with Doppler acting as a soft tiebreaker. When either side lacks valid v_r , the cost reverts to c_{geo} alone.

Near-zero v_r as a valid cue. Objects with $v_r \approx 0$ (e.g., laterally-moving cars) still produce consistent Doppler scores because the velocity difference Δv_r remains small. Filtering out such pairs degrades performance, indicating that near-zero v_r carries valid matching information.

C. Score-Adaptive Measurement Noise (SAN)

Detection confidence can serve as a practical proxy for localization reliability: high-score detections tend to correspond to well-localized objects, while low-score detections are more spatially uncertain. Standard Kalman filters apply a fixed measurement noise covariance $\mathbf{R}$, treating all detections equally regardless of confidence. The score is exploited as a proxy for localization reliability by scaling $\mathbf{R}$ adaptively (Fig. 3). Given score $s \in [0, 1]$:

$$\lambda = \max(1 - k \cdot s, \lambda_{\min}) \quad (5)$$

where $k = 0.7$ is the score sensitivity and $\lambda_{\min} = 0.3$. This yields $\lambda \in [0.3, 1.0]$: a high-confidence detection ($s \rightarrow 1$) yields $\lambda = 0.3$, reducing $\mathbf{R}$ and directing the filter to trust the measurement more; a low-confidence detection ($s \rightarrow 0$) yields

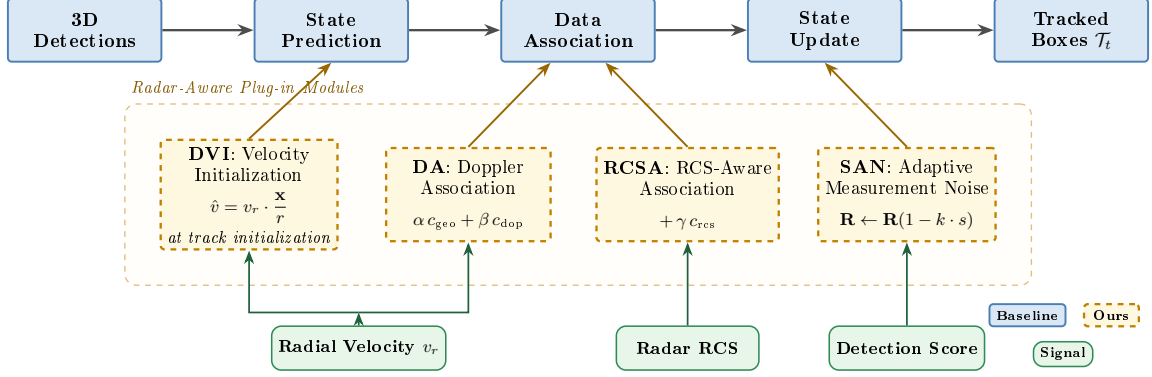


Fig. 1. Overview of DE-FastPoly. Four plug-in modules inject radar cues into the predict-associate-update cycle: Doppler velocity initialization (DVI), measured- v_r , Doppler association (DA), score-adaptive measurement noise (SAN), and optional RCS association (RCSA, Ped-only).

Fig. 2. Class-wise Distribution of Radial Velocity on VoD

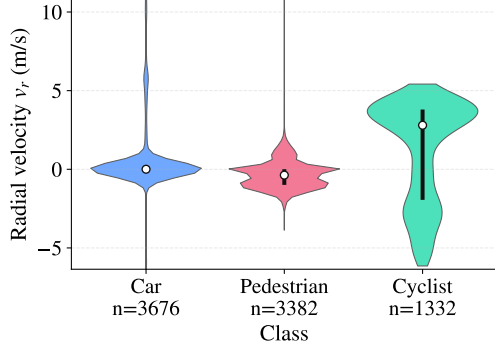


Fig. 2. Per-class distribution of radial velocity (v_r) on VoD (violin plot). Cars often exhibit near-zero v_r , corresponding to near-lateral motion relative to the sensor; Cyclists show a large spread. These class-dependent Doppler characteristics motivate velocity-aware association.

Fig. 3. Score-Adaptive Measurement Noise Scaling

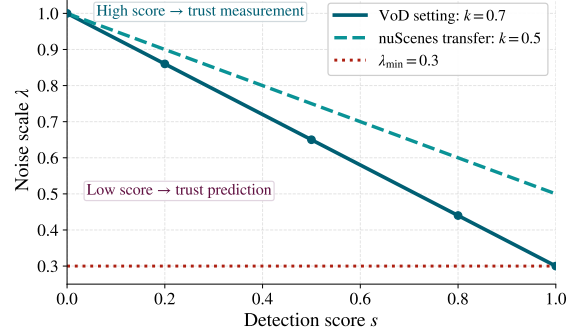


Fig. 3. Score-adaptive noise scaling. $\lambda = \max(1 - ks, \lambda_{\min})$. Higher detection score reduces $\mathbf{R}$, directing the Kalman filter to trust the measurement more.

$\lambda = 1.0$, preserving the default noise level. In the radar-based detection setting, the detector score provides a practical proxy for measurement reliability, as higher-confidence detections tend to correspond to objects with denser and more consistent radar returns.

Position-only scaling. A critical design choice is to scale only the position dimensions of $\mathbf{R}$:

$$R_{xx} \leftarrow \lambda \cdot R_{xx}, \quad R_{yy} \leftarrow \lambda \cdot R_{yy} \quad (6)$$

where R_{xx} and R_{yy} are the diagonal position entries of measurement noise covariance $\mathbf{R}$, leaving orientation and other dimensions unchanged. Empirically, extending adaptive scaling to all observed dimensions degrades tracking; this is attributed to interference with the indirect velocity estimation derived from position innovations, as well as disruption of well-calibrated orientation noise (see Table II for the full- $\mathbf{R}$ vs. pos-only comparison). The parameters k and $\lambda_{\min}$ are evaluated in the sensitivity study (Section IV-D); MOTA varies by less than 0.005 across $\pm 50\%$ perturbations.

D. RCS-Aware Association (RCSA)

Radar cross section (RCS) measures the amount of radar energy reflected by an object and can provide a complementary reflectivity-based cue for data association. RCS consistency is explored as a preliminary, training-free soft additive cue in the association cost. This module is presented as a dataset-specific enhancement rather than a universally applicable contribution; its utility is limited to scenarios where RCS measurements are stable and class-consistent.

RCS measurement and storage. For each detection, RCS^{det} is computed as the mean RCS of radar points falling inside the detection bounding box. Each track stores RCS^{trk} , the RCS value of its most recently matched detection. If either the detection or the track lacks valid RCS, the term is disabled for that pair.

RCS consistency cost. For each detection-track pair where both carry valid RCS values, the cost is computed as:

$$c_{\text{rcs}} = 1 - \exp\left(-\frac{(\Delta \text{RCS})^2}{2\sigma_{\text{rcs}}^2}\right) \quad (7)$$

where $\Delta\text{RCS} = \text{RCS}^{\text{det}} - \text{RCS}^{\text{trk}}$ and $\sigma_{\text{rcs}} = 6.0 \text{ dBsm}$. The final cost adds this as a soft term to c from Eq. (4):

$$c_{\text{final}} = c + \gamma \cdot c_{\text{rcs}}, \quad \gamma = 0.005 \quad (8)$$

The small weight γ ensures RCS acts as a soft cue rather than a hard constraint, since RCS measurements can be noisy in automotive environments.

Gating conditions. Two conditions must be satisfied for the RCS term to be applied: (1) Score gate: detection score ≥ 0.3 ; low-confidence detections often enclose noisy radar returns with unreliable RCS estimates. (2) Class filter: RCS is applied only to Pedestrians. For Cars, aspect-angle-dependent reflectivity causes large RCS variation across frames, making it an unreliable matching signal. Empirically, on VoD, Pedestrian RCS is more stable across consecutive frames than Car RCS.

Note that RCSA is treated as a **scene-specific exploratory enhancement** rather than a core generalizable contribution. Its effectiveness is evaluated on VoD Pedestrian tracks (Section IV-C), and its applicability to other datasets is left for future investigation.

IV. EXPERIMENTS

A. Experimental Setup

Evaluation is conducted primarily on the View-of-Delft (VoD) dataset [3], which provides synchronized 4D imaging radar, LiDAR, and camera data for urban driving, using the official validation split (Car, Pedestrian, Cyclist). For cross-dataset transfer, nuScenes [10] (validation split) is used. VoD detections use PointPillars [11] with score threshold 0.2; nuScenes uses CenterPoint [12]. All compared methods share the same detector outputs on VoD. On VoD, MOTA, MOTP, and IDS are reported; MOTA is defined as $1 - (\text{FP} + \text{FN} + \text{IDS})/N_{\text{GT}}$. On nuScenes, the official protocol is followed, reporting average MOTA (AMOTA), AMOTP, MOTA, and IDS.

Oracle setting. To isolate the contribution of the tracking modules from detection quality, all ablation configurations are additionally evaluated using ground-truth boxes as detections. This oracle setting eliminates detection-level noise, allowing cleaner assessment of association and state estimation behavior. In the oracle setting, GT boxes are assigned a fixed score of 0.9 for use by the SAN module and the RCSA score gate.

Implementation. For the FastPoly baseline and DE-FastPoly, track retention is set to $\text{max_age} = 2$ for all classes. This choice is motivated by the static-sensor VoD setup, where long retention accumulates ghost tracks more aggressively than in ego-motion-heavy benchmarks such as nuScenes; $\text{max_age} = 2$ reduces spurious track persistence while preserving association continuity. All remaining hyperparameters follow the FastPoly defaults. The implementation will be released after acceptance.

Compared methods. DE-FastPoly is compared against Poly-MOT [6], ByteTrack [7] (3D-adapted), and SimpleTrack [2], representative tracking-by-detection baselines that do not exploit radar-specific cues. All methods use the same PointPillars detections, evaluation script, and $\text{max_age} = 2$.

TABLE I
MAIN RESULTS ON VoD (VAL, POINTPILLARS, SCORE ≥ 0.2). BOLD: BEST PER COLUMN.

Method	MOTA $\uparrow$	MOTP $\downarrow$	IDS $\downarrow$	TP $\uparrow$	FP $\downarrow$	FN $\downarrow$
FastPoly (baseline)	0.294	0.347	169	4052	1095	5422
Poly-MOT †	0.301	0.268	290	4927	1786	4547
ByteTrack †	0.321	0.268	417	3888	433	5586
SimpleTrack	0.345	0.245	151	4656	1239	4818
DE-FastPoly	0.361	0.265	127	4450	907	5024

† Re-implemented / adapted to VoD 3D protocol.

TABLE II
ORACLE ABLATION ON VoD (GT BOXES AS DETECTIONS). BOLD: BEST PER COLUMN.

Config	MOTA $\uparrow$	IDS $\downarrow$	TP $\uparrow$	FP $\downarrow$	FN $\downarrow$
Baseline	0.837	190	8521	404	953
DVI	0.848	171	8618	411	856
DA	0.891	153	8907	312	567
DVI+DA	0.892	148	8912	313	562
DVI+DA+SAN (full-R)	0.917	140	9056	228	418
DVI+DA+SAN (pos-only)	0.953	59	9211	122	263
DVI+DA+SAN (pos)+RCSA	0.954	41	9207	125	267

ByteTrack and Poly-MOT are re-implemented under the VoD 3D protocol (3D Euclidean cost, BEV overlap). DE-FastPoly additionally exploits radar-derived attributes available alongside the detections. RaTrack [8] requires task-specific training and is not comparable under the learning-free protocol used here.

B. Main Results on VoD

Table I reports results on the VoD validation set. DE-FastPoly achieves MOTA 0.361, a +6.7 percentage-point improvement over the FastPoly baseline (0.294) while reducing identity switches from 169 to 127. Among compared methods, SimpleTrack reaches the closest MOTA (0.345) but produces more identity switches (151 vs. 127), indicating that DE-FastPoly achieves stronger identity consistency. Poly-MOT achieves MOTA 0.301 with 290 identity switches under the same protocol, substantially more IDS than DE-FastPoly (127), highlighting the benefit of Doppler-aware association for identity consistency. DE-FastPoly runs at 216.8 FPS, maintaining real-time performance with small additional latency compared to the FastPoly baseline (see Section IV-D).

C. Ablation Study

Tables II and III present oracle and detector-based ablations (DVI: Doppler init; DA: Doppler assoc.; SAN: score-adaptive noise; RCSA: RCS assoc.).

Oracle setting (Table II). Using ground-truth boxes isolates the tracker’s contribution. DVI alone improves MOTA by +1.1 pp (0.848) by providing better velocity initialization at track birth; the marginal FP increase (404→411) reflects that improved trajectory prediction occasionally retains borderline tracks longer. DA alone yields a larger +5.4 pp gain (0.891), indicating that Doppler-consistency association is the stronger of the two modules under clean detection. Their combination

TABLE III
DETECTOR ABLATION ON VoD (POINTPILLARS, SCORE $\geq$ 0.2). BOLD:
BEST PER COLUMN.

Config	MOTA $\uparrow$	IDS $\downarrow$	TP $\uparrow$	FP $\downarrow$	FN $\downarrow$
Baseline	0.294	169	4052	1095	5422
DVI+DA	0.312	139	4211	1118	5263
DVI+DA+SAN (full-R)	0.322	148	4305	1111	5169
DVI+DA+SAN (pos-only)	0.358	136	4437	905	5037
DVI+DA+SAN (pos)+RCSA	0.361	127	4450	907	5024

DVI+DA (+5.5 pp, MOTA 0.892) shows that DA is the dominant contributor under oracle detection, while DVI provides a small complementary gain at track birth.

Adding SAN with full-R scaling brings a further +2.5 pp (MOTA 0.917, IDS 140). However, switching to position-only scaling yields a substantially larger gain: +6.1 pp over DVI+DA (MOTA 0.953, IDS 59). The IDS drop from 140 to 59 under position-only scaling indicates that constraining the scale to position dimensions preserves well-calibrated orientation noise and avoids disrupting the indirect velocity estimation (see Section III-C). Finally, adding RCSA as a pedestrian-specific soft cue reduces IDS further to 41 (MOTA 0.954); the small FP and FN increases (122 $\rightarrow$ 125, 263 $\rightarrow$ 267) reflect occasional over-association via the RCS cue, outweighed by the identity-consistency gain.

Detector setting (Table III). Under realistic detection, DVI+DA improves MOTA by +1.8 pp; the slight FP increase (1095 $\rightarrow$ 1118) reflects that the Doppler soft term retains some marginal associations that the geometric-only baseline discards. Consistent with the oracle finding, full-R scaling (MOTA 0.322) underperforms position-only scaling (MOTA 0.358) and increases identity switches relative to DVI+DA alone (148 vs. 139), as full-R scaling disrupts the indirect velocity estimation, degrading identity consistency. Adding SAN (position-only) adds a further +4.6 pp over DVI+DA, indicating that score-adaptive noise scaling contributes the largest additional MOTA gain among the tested modules under detector noise. The RCSA module contributes an additional marginal improvement (+0.3 pp MOTA, -9 IDS), consistent with its role as a scene-specific soft cue; the slight FP increase (905 $\rightarrow$ 907) reflects rare over-association of nearby pedestrian detections via the RCS cue. Compared with the oracle setting, the overall gains are smaller under realistic detection because detector-induced false positives and missed detections partially mask the benefit of improved association and state estimation.

Per-class (Table IV). Pedestrian tracking benefits most: MOTA improves from 0.177 to 0.284 (+10.7 pp) and IDS drops from 83 to 55 (-34%). This large gain is driven by the combination of score-adaptive noise scaling and Doppler-aware association, both of which are particularly beneficial for low-confidence, motion-ambiguous classes. Cyclist MOTA improves by +7.1 pp with IDS reducing from 25 to 16 (-36%). Car gains are smaller (+2.9 pp, IDS -8%), as Cars already have high detection scores and well-localized returns, leaving less room for improvement.

TABLE IV
PER-CLASS BREAKDOWN (DETECTOR SETTING, BASELINE VS.
DE-FASTPOLY).

Class	BL MOTA $\uparrow$	Ours MOTA $\uparrow$	BL IDS $\downarrow$	Ours IDS $\downarrow$
Car	0.326	0.355 (+2.9)	61	56 (-8%)
Pedestrian	0.177	0.284 (+10.7)	83	55 (-34%)
Cyclist	0.506	0.577 (+7.1)	25	16 (-36%)
Overall	0.294	0.361 (+6.7)	169	127 (-25%)

TABLE V
TRACKING LATENCY ON VoD (CORE TRACKING, PRE-LOADED
DETECTIONS, 8 RUNS).

Config	ms/frame $\downarrow$	FPS $\uparrow$	Overhead $\downarrow$
Baseline	4.34	230.5	—
DVI+DA	4.44	225.4	+2.2%
DVI+DA+SAN (pos-only)	4.54	220.5	+4.5%
DVI+DA+SAN (pos)+RCSA	4.61	216.8	+6.3%

TABLE VI
PARAMETER SENSITIVITY (VoD ORACLE, $\pm$ 50% PERTURBATION).
MOTA $\uparrow$ SHOWN. MAX $|\Delta|$ IS COMPUTED FROM UNROUNDED VALUES.

	σ	α	k	$\lambda_{\min}$	γ
Nominal	10.0	0.95	0.7	0.3	0.005
-50%	0.955	0.952	0.950	0.954	0.954
Nominal	0.954	0.954	0.954	0.954	0.954
+50%	0.953	0.955	0.955	0.953	0.955
Max $ \Delta $	0.002	0.003	0.004	0.001	0.001

D. Runtime and Parameter Sensitivity

Runtime. Table V reports tracking latency (pre-loaded detections, core tracking only; averaged over 8 runs on an Intel i7-13700K, Python 3.11). The FastPoly baseline runs at 4.34 ms/frame (230.5 FPS). The full DE-FastPoly method (all four modules) runs at 4.61 ms/frame (216.8 FPS), an overhead of +6.3% relative to the baseline. All four modules involve only scalar operations per detection-track pair (exponential functions, element-wise matrix scaling), with no matrix decompositions or neural network inference, keeping the overhead small. All configurations maintain real-time performance at over 200 FPS.

Parameter sensitivity. Table VI reports robustness to the five key hyperparameters on the VoD oracle setting; the oracle is used here to isolate the tracker’s internal sensitivity from detector noise, providing a cleaner signal for hyperparameter analysis. Each parameter is independently perturbed by $\pm$ 50% from its nominal value. Across σ (Doppler noise, 10.0 m/s), α (blend factor, 0.95), k (SAN score sensitivity, 0.7), $\lambda_{\min}$ (SAN floor, 0.3), and γ (RCSA weight, 0.005), the maximum MOTA change is 0.004 (at k -50%). All other parameters produce Δ MOTA $\leq$ 0.003. The small variation across all tested parameters indicates the proposed modules are robust to perturbation on VoD; cross-dataset transfer (Section IV-E) requires only minor adaptation.

TABLE VII
CROSS-DATASET TRANSFER ON nuSCENES (VAL, CENTERPOINT
DETECTIONS). BOLD: BEST PER COLUMN.

Method	AMOTA $\uparrow$	AMOTP $\downarrow$	MOTA $\uparrow$	IDS $\downarrow$
FastPoly (baseline)	0.7367	0.5301	0.6319	414
DE-FastPoly (DA+SAN)	0.7367	0.5287	0.6324	347

E. Cross-Dataset Transfer (nuScenes)

DE-FastPoly is evaluated on nuScenes as a transfer generalizability check rather than a primary benchmark. Two dataset-specific adaptations are made: (1) module DVI is disabled because the CenterPoint detector already provides per-detection velocity estimates, making Doppler initialization redundant; (2) SAN score sensitivity is reduced to $k=0.5$ (vs. $k=0.7$ on VoD) to reflect the different detector score distribution. No other parameters are changed. For module DA, where per-detection radar radial velocity is not directly available on nuScenes, the radial component of CenterPoint’s per-detection velocity projected onto the sensor ray is used, providing an analogous velocity-consistency cue rather than a direct Doppler measurement.

Table VII reports results on the nuScenes validation set. The DA+SAN configuration maintains AMOTA (0.7367 vs. 0.7367) and MOTA (0.6324 vs. 0.6319) while reducing IDS by 67 (−16%). The smaller MOTA improvement compared to VoD is expected: CenterPoint already provides strong per-detection velocity priors, making module DVI unnecessary in this setting, and reducing the marginal impact of velocity-consistency association. The consistent IDS reduction without AMOTA degradation provides initial evidence of transferability; a more complete evaluation on additional 4D radar benchmarks is left for future work.

F. Qualitative Analysis

Fig. 4 presents a camera-view comparison from the VoD detector setting with a unified 2×3 layout per case.

Success case. A Pedestrian (GT 773, $v_r=+0.75$ m/s) around frame 199. The baseline tracker incurs an identity switch (T103 $\rightarrow$ T110), while DE-FastPoly keeps a stable identity (T80) across frames 197–199. This visual pattern is consistent with the dataset-level IDS reduction reported in Tables I and III, indicating that Doppler-aware association improves identity consistency under detector noise.

Failure case. A Pedestrian (GT 21, $v_r=-1.08$ m/s) around frame 8377 where both methods incur an identity switch (Baseline: T34 $\rightarrow$ T39; DE-FastPoly: T35 $\rightarrow$ T37). This indicates that Doppler-aware association alone is insufficient when the radial-velocity contrast between competing tracks is limited, and complements the quantitative results by showing that DE-FastPoly reduces IDS overall but does not eliminate all failure modes.

V. CONCLUSION

This paper has presented DE-FastPoly, a plug-in extension to the FastPoly tracker that incorporates radar-native

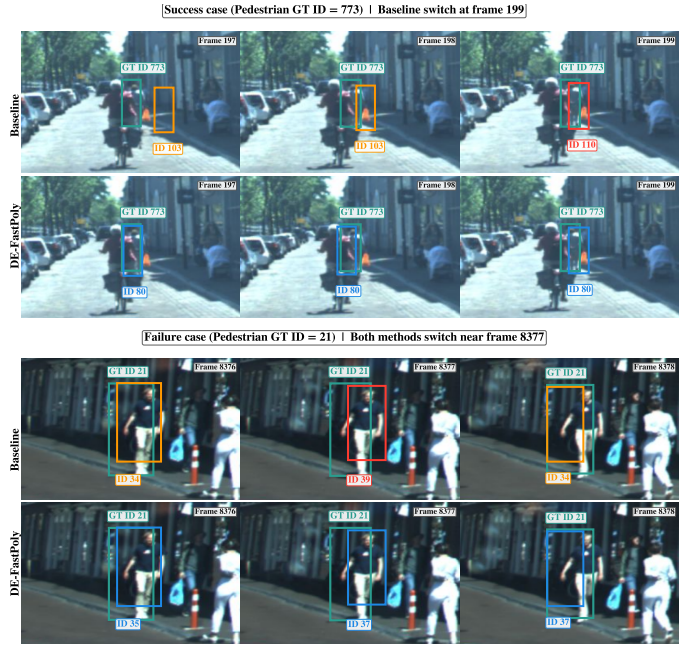


Fig. 4. Camera-view qualitative comparison on VoD (detector setting). Top: success case (GT 773, frames 197–199). Bottom: failure case (GT 21, frames 8376–8378). Green: GT box; orange/red: baseline (pre/post-switch ID); blue: DE-FastPoly. See text for analysis.

attributes into a learning-free multi-object tracking pipeline. Three core modules address complementary stages of the predict-associate-update cycle: Doppler velocity initialization (DVI), measured- v_r association (DA), and score-adaptive measurement noise (SAN). An optional RCS-aware association module (RCSA) provides a scene-specific pedestrian-tracking enhancement. On the VoD dataset, the full method improves MOTA by 6.7 pp over the FastPoly baseline under realistic detection and by 11.7 pp in the oracle setting, while maintaining real-time throughput (217 FPS, +6.3% latency). Cross-dataset transfer on nuScenes with a DA+SAN configuration yields a 16% IDS reduction without AMOTA degradation, providing initial evidence that the velocity-consistency and score-adaptive noise modules transfer across sensor configurations with only minor adaptation.

The RCSA module is validated only on VoD Pedestrian tracks; future work includes Doppler-based velocity observation in the Kalman state and evaluation on additional 4D radar datasets.

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